

Homework4

1. Calculate the following points

$$(1) \int_0^{2\pi} \cos^{2n} \theta d\theta, \quad n \in \mathbb{Z}_{\geq 1}$$

$$(3) \int_0^{\pi} \frac{d\theta}{1 + \sin^2 \theta};$$

$$(2) \int_0^{2\pi} \frac{dx}{(a + b \cos x)^2}, \quad a > b > 0;$$

$$(4) \int_0^{\pi} \frac{d\theta}{(1 + \sin^2 \theta)^2}.$$

2. Calculate the following points

$$(1) \int_{-\infty}^{\infty} \frac{x^2}{1 + x^4} dx;$$

$$(3) \int_{-\infty}^{\infty} \frac{x^{2m}}{1 + x^{2n}} dx, \quad n, m \in \mathbb{Z} \text{ \& } n > m$$

$$(4) \int_{-\infty}^{\infty} \frac{dx}{(1 + x^2) \cosh \frac{\pi x}{2}}.$$

$$(2) \int_{-\infty}^{\infty} \frac{1}{(1 + x^2)^{n+1}} dx, \quad n \in \mathbb{Z}_{\geq 1}$$

3. Calculate the following points

$$(1) \int_0^{\infty} \frac{\cos x}{1 + x^4} dx;$$

$$(3) \int_{-\infty}^{\infty} \frac{x \sin x}{x^2 - 2x + 2} dx;$$

$$(4) \int_0^{\infty} \frac{\sin(a + 2n)x - \sin ax}{(1 + x^2) \sin x} dx, \quad a > -1, \quad n \in \mathbb{Z}_{\geq 1}$$

$$(2) \int_0^{\infty} \frac{\cos x}{(1 + x^2)^3} dx;$$

4. Calculate the following points

$$(1) \text{v.p.} \int_{-\infty}^{\infty} \frac{dx}{x(x-1)(x-2)};$$

$$(3) \int_0^{\infty} \frac{x - \sin x}{x^3(1 + x^2)} dx;$$

$$(2) \int_0^{\infty} \frac{\sin(x+a) \sin(x-a)}{x^2 - a^2} dx, \quad a > 0;$$

$$(4) \int_{-\infty}^{\infty} \frac{e^{px} - e^{qx}}{1 - e^x} dx, \quad 0 < p < 1, \quad 0 < q < 1.$$